

Discussion: Theoretical Implications, Limitations, and Future Directions

This section examines the theoretical implications of the Cognitive Integrity Framework (sec:theoretical-implications), formal limitations and boundary conditions (sec:limitations), relationship to prior work (sec:related-work), governance implications (sec:governance), and future research directions (sec:future-directions).

Empirical Validation Summary

The formal mechanisms proposed in this paper have been validated through extensive architecture-aware parametric simulation in **Part 2: Computational Validation**. In experiments across six production multiagent architectures (including hierarchical, role-based, and peer-to-peer topologies), the full Cognitive Integrity Framework achieved a **94% overall detection rate** against a corpus of 950 cognitive attacks.

Key empirical findings include:

1. **Defense Composition:** Consistent with Theorem 3.1 and 3.2, the full defense mechanism consistently outperforms individual components, achieving the full